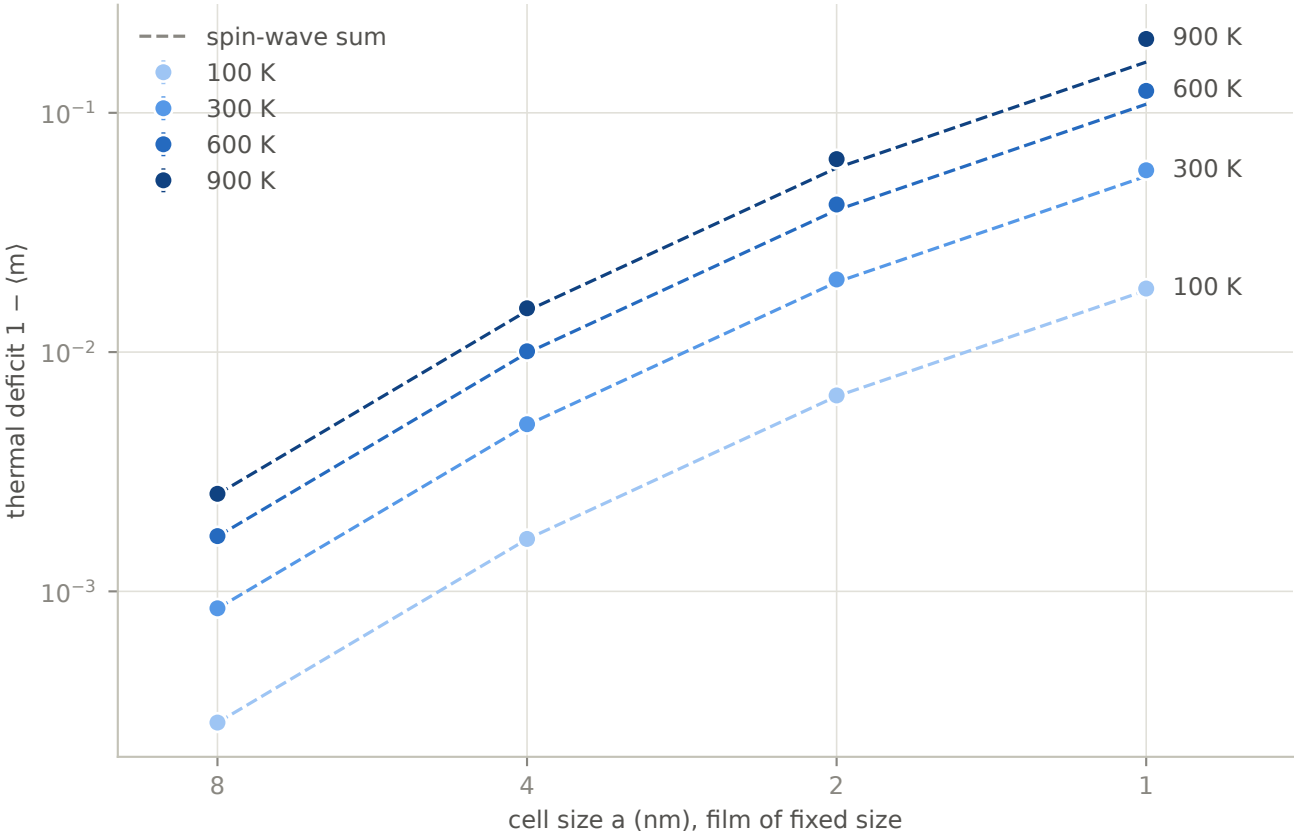
Thermal results against cell size

Finer meshes make the same film look hotter, and the result does not converge. At 300 K the thermal deficit grows from 0.00085 at 8 nm to 0.0575 at 1 nm. Each halving of the cell multiplies it by 5.9, then 4.0, then 2.9, and the spin-wave sum heads for 2. The sum holds at low T: the measured deficit runs 2% above it at a 2% deficit, and 25% above it at a 20% deficit.



a (nm)	cells	T (K)	⟨m⟩ ± σ	1 − ⟨m⟩	sum	measured / sum	⟨m⟩ at dt/2
8	16	100	0.9997171 ± 0.0000001	0.00028	0.00028	1.001	—
8	16	300	0.9991504 ± 0.0000003	0.00085	0.00085	1.002	0.9991503 ± 0.0000003
8	16	600	0.9982982 ± 0.0000006	0.00170	0.00170	1.003	—
8	16	900	0.9974432 ± 0.0000009	0.00256	0.00254	1.005	0.9974426 ± 0.0000009
4	128	100	0.9983437 ± 0.0000006	0.00166	0.00165	1.002	—
4	128	300	0.9950039 ± 0.0000018	0.00500	0.00496	1.008	0.9950015 ± 0.0000015
4	128	600	0.9899234 ± 0.0000038	0.01008	0.00992	1.016	—
4	128	900	0.9847531 ± 0.0000059	0.01525	0.01487	1.025	0.9847430 ± 0.0000045
2	1024	100	0.9934107 ± 0.0000027	0.00659	0.00654	1.008	—
2	1024	300	0.9798844 ± 0.0000078	0.02012	0.01961	1.026	0.9798826 ± 0.0000067
2	1024	600	0.9586167 ± 0.0000154	0.04138	0.03921	1.055	—
2	1024	900	0.9359509 ± 0.0000234	0.06405	0.05882	1.089	0.9359006 ± 0.0000254
1	8192	100	0.9815663 ± 0.0000041	0.01843	0.01811	1.018	—
1	8192	300	0.9424693 ± 0.0000124	0.05753	0.05432	1.059	0.9424648 ± 0.0000309
1	8192	600	0.8764933 ± 0.0000310	0.12351	0.10864	1.137	—
1	8192	900	0.7964515 ± 0.0000629	0.20355	0.16296	1.249	0.7961610 ± 0.0001279

Film 32 × 32 × 8 nm, periodic in x and y, free in z, no demag, H = 0, m starts along +z. $Nd_2Fe_{14}B$: $\mu_0 M_s$ = 1.61 T, K = 4.30 MJ/m³, A = 7.7 pJ/m, easy axis z. Heun, α = 1, dt = 1/160 of the fastest spin-wave period. 8 seeds per point, each 0.25 ns · (a/nm)³ after a 20 ps burn-in; σ from the spread of the seed means. Halving dt again moves the deficit by at most 0.14% (2σ).
Why the slope changes: well below $\sqrt{(A/K)}$ = 1.3 nm, exchange ties neighbouring cells and the deficit goes as 1/a; far above it, each cell acts as its own macrospin and the deficit goes as $1/a^3$. Real $Nd_2Fe_{14}B$ has T_c = 585 K, so 600 and 900 K probe the model, not the material. Decision D52. Regenerate with benchmarks/thermal_cell_size.py.